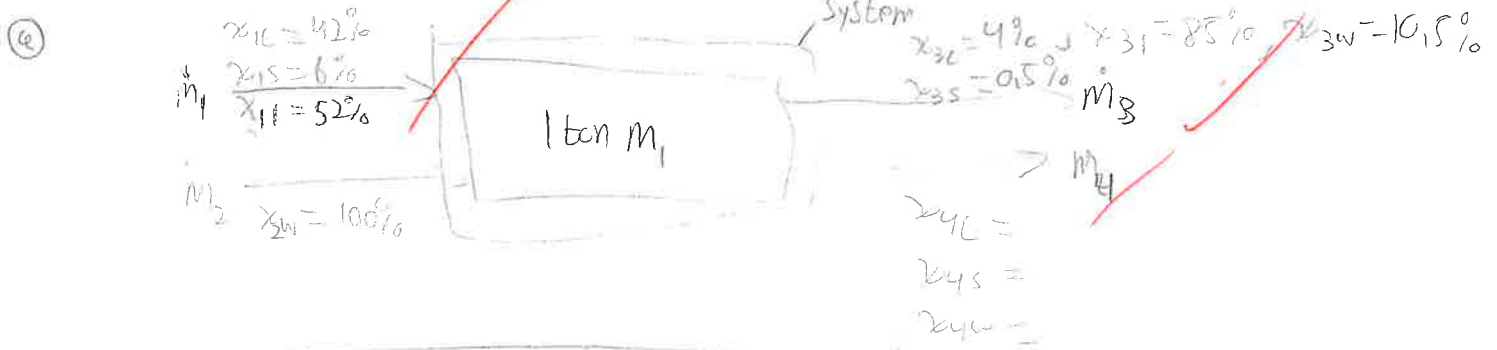


To Receive Full Credit, You Must Show All Your Work

The LeBlanc process produces "black ash" which is a solid consisting of 42 wt% sodium carbonate (C), 6 wt% water-soluble material (S), and 52 wt% water-insoluble material (I). The black ash is treated with liquid water (W) to extract the sodium carbonate. The result of the process produces a solid residue containing: 4 wt% C, 0.5 wt% S, 85 wt% I, and 10.5 wt% W. Also, a liquid product containing C, S, and W is produced. Use a basis of $m_1 = 1$ ton of black ash. Label the mass of liquid water added as m_2 , the mass of the solid residue produced as m_3 , and the mass of the liquid product m_4 . Give all your answers to the tenth of a lb_m .

- (a) Draw and label a block flow diagram. Define a system.
(b) Calculate the mass of the solid residue, m_3 , the mass of the sodium carbonate extracted to the liquid product, $m_{4,C}$, and the mass of S extracted to the liquid product, $m_{4,S}$, in lb_m .
(c) If the extraction process requires that the liquid product to be 75 wt% water, what is the mass in lb_m of the required liquid water added to the black ash, that is, m_2 , and the mass of the liquid product, m_4 ?



(b) DOF

4a 11 ✓ S_u 1 ✓
4b 0 ✓ S_{in} 1 ✓
11 - 11 = 0 ✓
 S_{in} 10 ✓
 S_{in} 14 ✓
 T_1

$$m_1 + m_2 = m_3 + m_4$$

Species	$m_1 (\text{lb})$	$m_2 (\text{lb})$	$m_3 (\text{lb})$	$m_4 (\text{lb})$
C	840 ✓	0	$0.04m_3$ ✓	$x_{4C}m_4$ ✓
S	120 ✓	0	$0.005m_3$ ✓	$x_{4S}m_4$ ✓
I	1040 ✓	0	$0.85m_3$ ✓	0 ✓
W	0	m_2	$0.105m_3$ ✓	$x_{4W}m_4$ ✓
total	2000 ✓	m_2	m_3 ✓	m_4 ✓

$$1040 = 0.85m_3$$

$$m_3 = 1223.53 \checkmark$$

$$x_{1C}m_1 = x_{3C}m_3 + x_{4C}m_4$$

$$x_{1S}m_1 = x_{3S}m_3 + x_{4S}m_4$$

$$x_{1I}m_1 = x_{3I}m_3$$

$$x_{2W}m_2 = x_{3W}m_3 + x_{4W}m_4$$

$$840 = 0m_2 + 0.04m_3 + x_{4C}m_4$$

$$120 = 0m_2 + 0.005m_3 + x_{4S}m_4$$

$$1040 = 0m_2 + x_{3I}m_3 + 0m_4$$

$$x_{3I} = 0.85$$

back

$$1040 = 0m_2 + x_{31}m_3 + 0m_4 \quad \text{--- From } x_{11}m_1 = x_{31}m_3$$

$$1040 = 0.185m_3$$

$$m_3 = 1223.53 \text{ lb} \quad \checkmark$$

$$C: 840 = 48.9 + x_{4c}m_4$$

$$S: 120 = 6.12 + x_{4s}m_4$$

$$I: 1040 = 1040$$

$$W: \cancel{x_{2w}m_2} = \cancel{0.105 \times m_3} + \cancel{x_{4w}m_4}$$

$$x_{2w}m_2 = 128.47 + x_{4w}m_4$$

$$791.1 \checkmark = x_{4c}m_4$$

$$113.88 \checkmark = x_{4s}m_4$$

$$x_{2w}m_2 - 128.47 = x_{4w}m_4$$

c) ————— 2